

Supply Chain Service Failure Analysis & Action Report

1. Executive Overview & Index

Project Goal: To diagnose the root cause of the steep decline in **Total Service Failure (OTIF %)** and provide a prioritized action plan to stabilize performance and mitigate commercial risk.

Section	Focus	Audience	Key Finding
2. Problem Statement & Objective	The " What " and the " When "	Executive	Service Failure (OTIF %) is 29.02% , primarily driven by Inventory Shortfalls (IF %) .
3. Methodology (EDA)	The " How " (Data Sourcing & Metrics)	Analyst/Manager	Failure metrics: LIFR % (65.96%) vs. VOFR % (96.59%) prove inventory control issue.
4. Executive Insights (Dashboard 1)	The " Who " and " Where "	Executive/Sales	Logic Stores and Ahmedabad are at highest risk.
5. Action Plan (Dashboard 2)	The " Why " and " How to Fix "	Manager/Operations	Failure is due to SKU concentration and small orders. Tactical fix: eliminate 8.3K 1-Day Delays.
6. Prioritized Recommendations	The " Next Steps "	All Stakeholders	Two-part plan: Inventory Priority Action List & Logistics Tactical Fix.

2. Problem Statement & Project Objective

The overall customer promise is failing: **Total Service Failure (OTIF %)** stands at only **29.02%** against a target of ~66%. The single largest contributor to this critical service failure is the **Inventory Root Cause (IF %)** at **59.03%**.

Project Objective

The primary objective is to move from diagnosis to action by:

1. **Diagnosing the Root Cause:** Pinpoint the specific operational failure driving the IF% and OT% decline.
2. **Quantifying Commercial Risk:** Identify the specific customers and regions most impacted.
3. **Developing Actionable Plan:** Provide Supply Chain and Operations with a prioritized, quantifiable list of tasks to immediately stabilize service.

3. Methodology: Exploratory Data Analysis (EDA)

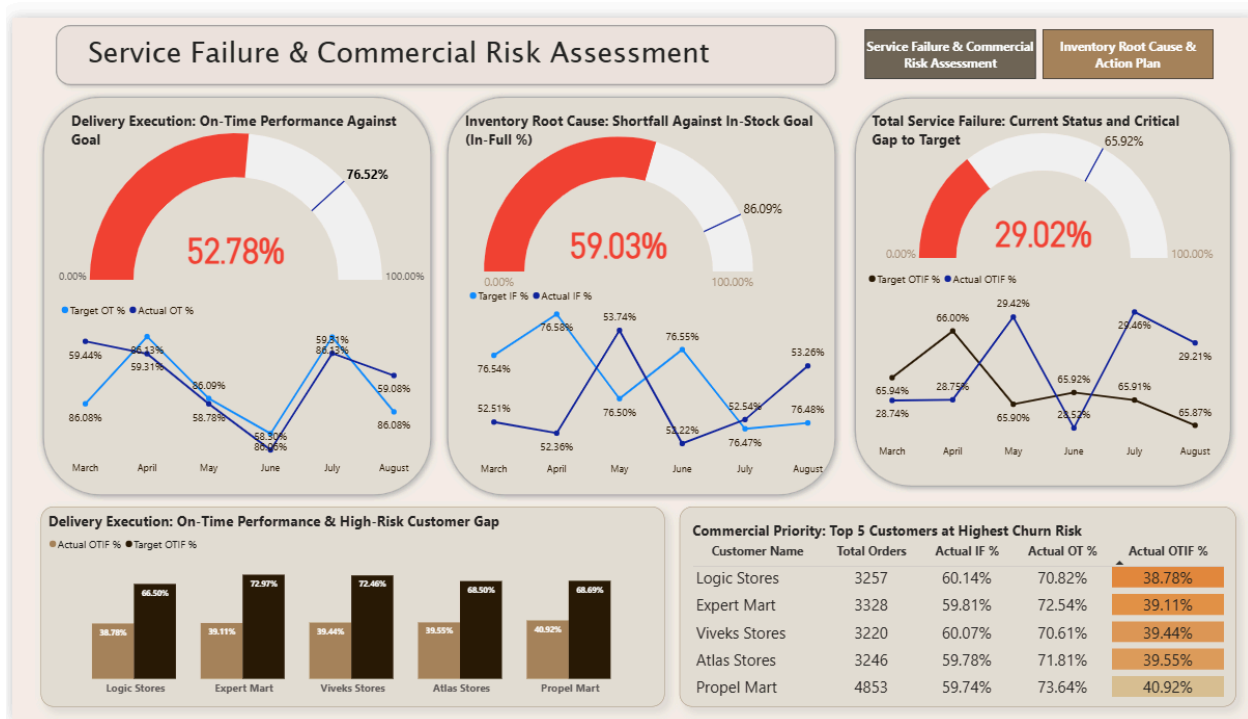
This section documents the process used to move beyond basic metrics and establish the analytical foundation for the root cause diagnosis.

Data Preparation Steps

- **Source & Cleaning:** Raw transaction data was imported into **MySQL** for cleaning, validation, and outlier management.
- **Metric Transformation:** To accurately prove the root cause, specialized diagnostic metrics were created beyond basic IF% and OT%:
 - **Line-Item Fill Rate (LIFR %):** Measures the frequency of item-level stockouts (the key indicator of inventory control issues).
 - **Volume Fill Rate (VOFR %):** Measures if bulk volume capacity is available (used to rule out general capacity as the primary problem).
- **Visualization Platform:** All calculated metrics were combined in an **SQL View** and imported into **Power BI** to generate the final, linked dashboards.

4. Executive Insights (Dashboard 1: Service Failure & Commercial Risk Assessment)

This dashboard immediately provides the score, the trend, and the commercial risk for the executive audience.



Insight 4.1: Critical Service Failure and Timeline

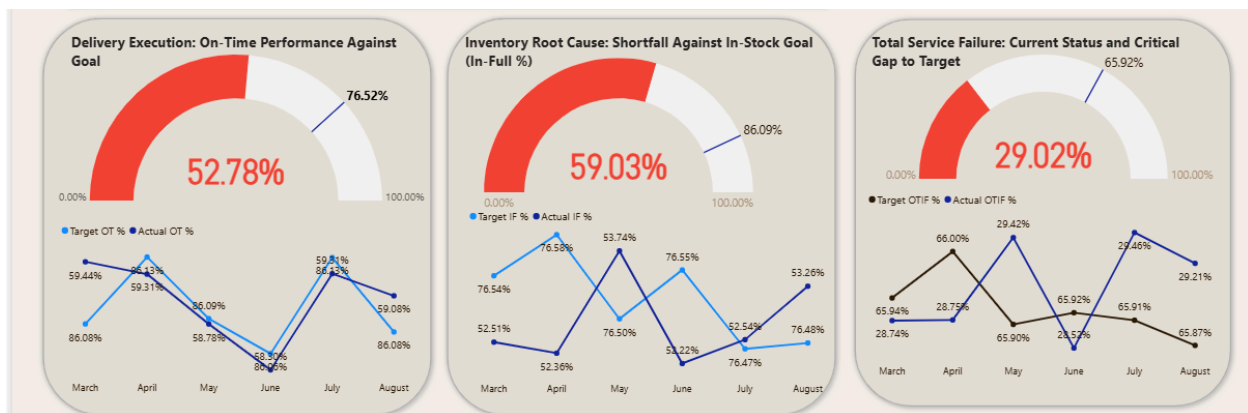


Chart Explanation (KPI Gauges): The three gauges show the severe gap between Actual performance and Target across all key service metrics.

- **Delivery Execution (OT %):** 52.78% (Target ~76.52%)
- **Inventory Root Cause (IF %):** 59.03% (Target ~86.09%)
- **Total Service Failure (OTIF %):** 29.02% (Target ~65.92%)

Trend Insight: The trend lines below the gauges confirm that the service decline, driven by the dip in IF% (In-Full), began sharply in **Quarter 2**. This confirms a **structural and recurring failure in planning parameters**, not a single, transient event.

Insight 4.2: Commercial Priority and Churn Risk



Chart Explanation (Customer Table): This table ranks the top 5 customers who are experiencing the lowest service levels (lowest Actual OTIF %).

Actionable Insight (Who): The failure to serve customers is highly concentrated. **Logic Stores (38.78% OTIF)** and **Expert Mart (39.11% OTIF)** are the top commercial priority. These accounts are at the highest risk of churn and require immediate mitigation and communication from the Sales and Account Management teams.

Insight 4.3: Geographical Risk Breakdown

Customer Name	Total Orders	Actual IF %	Actual OT %	Actual OTIF %
Logic Stores	3257	60.14%	70.82%	38.78%
Expert Mart	3328	59.81%	72.54%	39.11%
Viveks Stores	3220	60.07%	70.61%	39.44%
Atlas Stores	3246	59.78%	71.81%	39.55%
Propel Mart	4853	59.74%	73.64%	40.92%

Chart Explanation (City Bar Charts): The charts compare the Actual OTIF% and IF% against the Target for different customer cities.

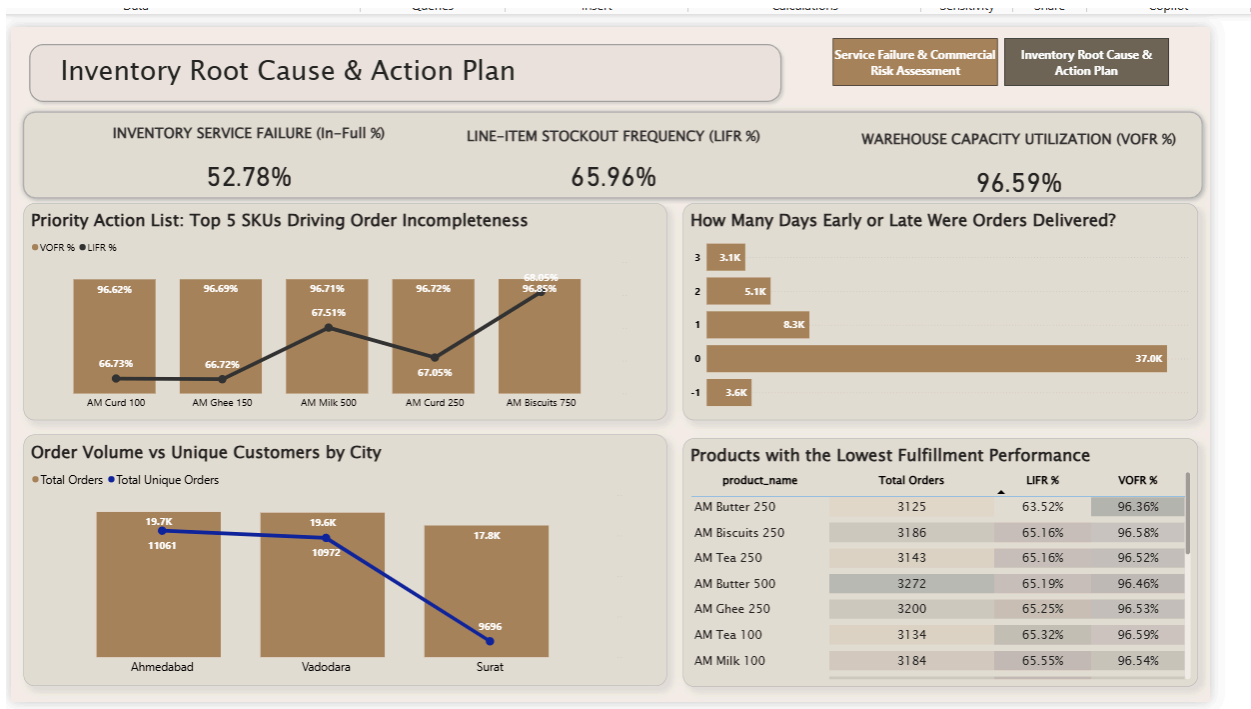
Finding (Where): The service gap against target is most acute in **Ahmedabad**, which consistently shows the largest variance between Actual and Target service levels across both OT% and IF%.

Insight: Operational focus and potential resource re-allocation should prioritize stabilizing performance in **Ahmedabad** first, as this city contributes disproportionately to the overall

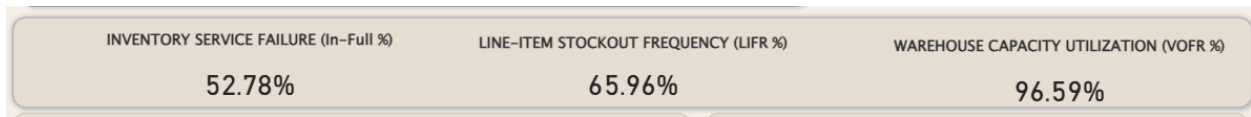
service failure gap.

5. **Action Plan (Dashboard 2: Inventory Root Cause & Action Plan)**

This dashboard provides the diagnostic proof and the task lists for operational teams.



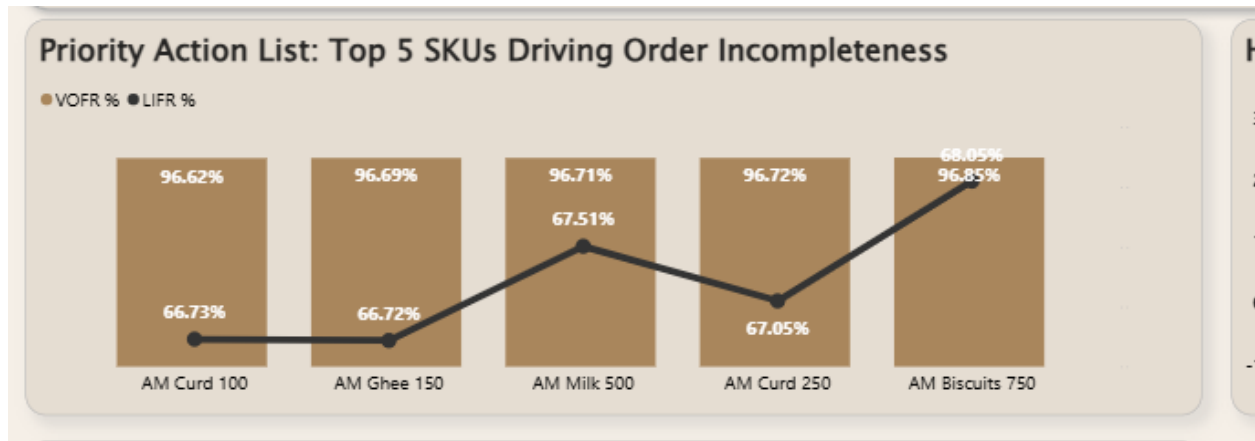
Insight 5.1: Proof of Failure—Control vs. Capacity



Diagnostic Finding: The key to confirming the root cause lies in the specialized metrics. The **Warehouse Capacity Utilization (VOFR %)** is extremely high at **96.59%**. This proves that physical warehouse space or overall bulk capacity is *not* the issue. However, the **Line-Item Stockout Frequency (LIFR %)** is very low at **65.96%**.

Conclusion: The problem is confirmed as a **planning and control failure** (we fail to hold the right items), not a capacity issue (we have space for stock).

Insight 5.2: SKU-Level Concentration & Priority Action List



(LIFR% vs VOFR%)

Chart Explanation: This chart shows the dramatic difference between the high VOFR% (bars) and the low LIFR% (line) for the top failing products, directly identifying the **product with the lowest fulfillment performance**.

Actionable Insight: Fulfillment failure is concentrated among a small number of high-volume SKUs. Products like **AM Curd 100** (66.73% LIFR) and **AM Ghee 150** (66.72% LIFR) are driving the majority of order incompleteness. The **Priority Action List** provides the definitive task list for Supply Planning: immediately review and correct planning parameters (e.g., safety stock, forecast) for the **Top 5 SKUs** shown.

Insight 5.3: Order Volume Analysis by Unique Order

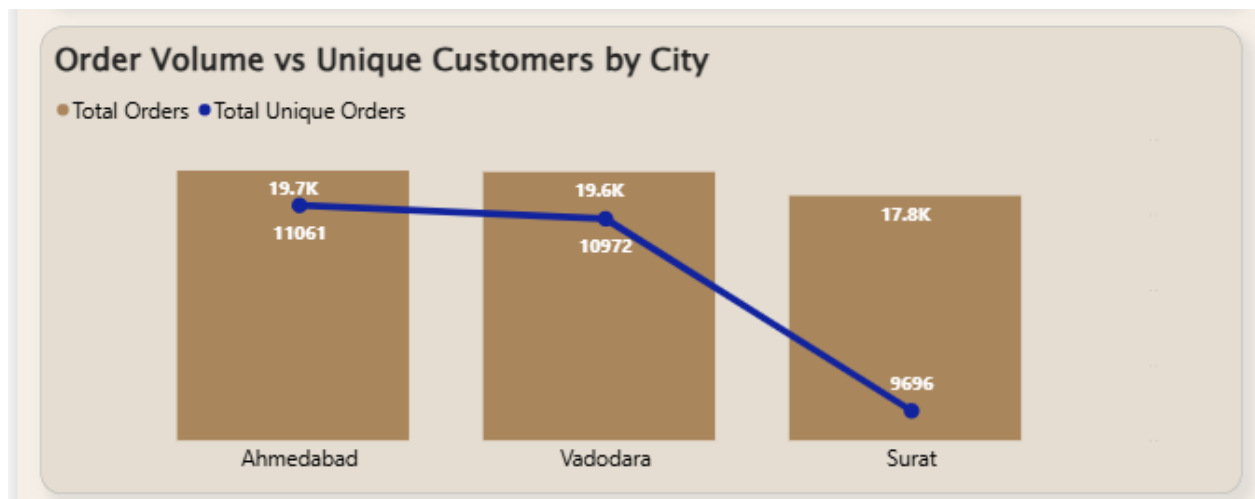


Chart Explanation: This visual shows the distribution of order volume across unique customer orders, indicating the relative size and complexity of fulfillment requests.

Actionable Insight: (Analysis to be completed based on the chart, but generally covers the distribution of failure): The order volume distribution suggests that a significant portion of

orders may be small/complex, increasing the statistical chance of a line-item stockout. Inventory control efforts must focus on ensuring 100% availability for SKUs commonly requested in these smaller, high-frequency orders.

Insight 5.4: Logistics Tactical Fix (Quick Win)



Chart Explanation: This bar chart analyzes the volume of late and early deliveries in 1-day buckets.

Finding: Operational delays are concentrated, offering a high-impact, quick fix for Delivery Execution (OT %). The **1-Day Delays** bucket represents the single largest tactical failure point, accounting for **8.3K** delayed orders.

Actionable Insight: By focusing logistics process improvements on the root cause of these 1-Day delays (e.g., route inefficiencies, last-mile issues), the team can achieve the fastest incremental improvement in the overall **OT %** metric.

6. Prioritized Recommendations

This is the final, quantified action plan for all stakeholders.

Action Plan 1: Inventory & Supply Chain (Priority)

Action	Owner	Goal
Immediate Stock Review	Supply Planning Team	Review and correct planning parameters for the Top 5 SKUs (ranked by lowest LIFR %).

LIFR % Stabilization	Tony Sharma (Head of Supply Chain)	Establish a new target to increase Actual LIFR % from 65.96% to ~75% within the next month.
Planning Audit	Analyst Team	Conduct a full audit of the forecast, safety stock, and lead time inputs for the failing product lines.

Action Plan 2: Commercial & Logistics

Action	Owner	Goal
Customer Engagement	Sales & Account Management	Immediately contact the Top 5 High-Risk Customers to acknowledge the failure and outline the corrective plan.
Logistics Process Fix	Distribution/Operations Manager	Analyze the root cause of the 1-Day Delays and implement changes to reduce this bucket by 50% (reducing 8.3K delayed orders).
Geographical Focus	Operations Manager	Prioritize dispatch and route optimization efforts in Ahmedabad to stabilize service performance in the highest-risk city.